

PiolinTech — WRO Future Engineers 2026 | Final Bill of Materials

Category	Qty.	Component	Specification / Model	Function	Connection / Location	Supplier	Part Number	Unit Price (USD)	Total Cost	Status	Notes
Control & Electronics	1	LEGO Mindstorms EV3	EV3 Brick	Main vehicle controller	Central chassis	Amazon (tracked)	LEGO 45500	\$209.84	\$209.84	Current	Amazon US tracked listing for EV3 Intelligent Brick.
Control & Electronics	1	EV3 Battery	Installed EV3 battery	Main EV3 power source	Inside EV3	Amazon (tracked)	LEGO 45501	\$130.76	\$130.76	Current	Amazon tracked listing for EV3 rechargeable battery.
Actuation	1	Drive Motor	LEGO motor	Rear-wheel propulsion	Motor Port A	Amazon (tracked proxy)	LEGO 45502	\$99.67	\$99.67	Current	Priced as EV3 Large Servo Motor replacement. Exact Motor A model should be verified on Piolin.
Actuation	1	Steering Motor	LEGO motor	Front Ackermann steering	Motor Port B	Amazon (tracked proxy)	LEGO 45503	\$93.56	\$93.56	Current	Priced as EV3 Medium Servo Motor replacement. Exact Motor B model should be verified on Piolin.
Vision Electronics	1	Arduino Nano	Nano-compatible board	Vision communication interface	USB to EV3	Amazon (tracked)	Arduino A000005	\$19.80	\$19.80	Current	Official Arduino Nano Amazon reference price. If Piolin uses a clone, actual cost may be lower.
Vision Electronics	1	USB Cable	Nano-to-EV3 cable	Vision data communication	Arduino Nano → EV3	Amazon / AmazonBasics proxy	USB-A to Mini-B 2.0	\$12.00	\$12.00	Current	Replacement USB data cable price reference; exact brand/cable length may differ.
Sensors	1	LEGO Ultrasonic Sensor	Front-facing	Frontal collision safety	EV3 S1	Amazon (tracked)	LEGO 45504	\$56.00	\$56.00	Current	EV3 Ultrasonic Sensor, front S1.
Sensors	1	LEGO Ultrasonic Sensor	Right lateral	Right-side wall measurement	EV3 S2	Amazon (tracked)	LEGO 45504	\$56.00	\$56.00	Current	EV3 Ultrasonic Sensor, right S2.
Sensors	1	LEGO Ultrasonic Sensor	Left lateral	Left-side wall measurement	EV3 S3	Amazon (tracked)	LEGO 45504	\$56.00	\$56.00	Current	EV3 Ultrasonic Sensor, left S3.
Sensors	1	LEGO Color Sensor	Downward-facing	Blue/orange floor-marking detection	EV3 S4	Amazon (tracked)	LEGO 45506	\$54.74	\$54.74	Current	EV3 Color Sensor.
Sensors	1	HuskyLens	Forward-facing vision sensor	Green/red pillar identification	Through Arduino Nano	Amazon-product reference	DfRobot SEN0305 / B085NL4XCB	\$46.08	\$46.08	Current	Standard HuskyLens v1 reference price; current availability can change.
Mobility	1	Rear Wheels	~61.0 mm diameter	Driven wheels	Rear drivetrain	Amazon/market replacement proxy	Technic wheel allocation	\$7.48	\$7.48	Current	Allocated replacement value for the two rear wheel rims. Exact installed rear-wheel SKU not confirmed.
Mobility	1	Front Wheels	38.1 mm diameter	Steering wheels	Front steering assembly	Amazon replacement proxy	Technic wheel allocation	\$2.90	\$2.90	Current	Allocated replacement value for the two front wheel rims. Exact installed front-wheel SKU not confirmed.
Mobility	1	Rear Tires	Compatible with rear wheels	Track traction	Rear wheels	Amazon/market replacement proxy	Technic tire allocation	\$7.48	\$7.48	Current	Allocated replacement value for the two rear tires. Exact installed tire SKU not confirmed.
Mobility	1	Front Tires	Compatible with front wheels	Steering contact with track	Front wheels	Amazon replacement proxy	Technic tire allocation	\$2.90	\$2.90	Current	Allocated replacement value for the two front tires. Exact installed tire SKU not confirmed.
Mobility	1	Rear Drivetrain Assembly	LEGO Technic	Transfers Motor A rotation to rear propulsion	Rear chassis	Amazon kit allocation	Habow 270pc Technic kit / B0C38JPBMD	\$8.00	\$8.00	Current	Allocated share of a \$19.99 Amazon-origin Technic gear/axle/steering parts kit.
Mobility	1	Ackermann Steering Linkage	LEGO Technic	Transfers Motor B movement to front wheels	Front chassis	Amazon kit allocation	Habow 270pc Technic kit / B0C38JPBMD	\$6.00	\$6.00	Current	Allocated share of the same \$19.99 steering/drivetrain parts kit.
Mobility	1	Front Steering Pivots / Hubs	LEGO Technic assembly	Allow front-wheel steering	Front chassis	Amazon kit allocation	Habow 270pc Technic kit / B0C38JPBMD	\$5.99	\$5.99	Current	Allocated share of the same \$19.99 steering/drivetrain parts kit.
Structure	1	LEGO Technic Beams	Various	Main chassis structure	Throughout chassis	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$4.50	\$4.50	Current	Allocated share of a \$24.90 Amazon-tracked Technic assortment.
Structure	1	LEGO Technic Frames	Various	Structural reinforcement	Throughout chassis	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$3.50	\$3.50	Current	Allocated share of the same \$24.90 Technic assortment.
Structure	1	LEGO Technic Pins	Various	Structural connections	Throughout chassis	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$3.00	\$3.00	Current	Allocated share of the same \$24.90 Technic assortment.
Structure	1	LEGO Technic Axles	Various	Wheel/drivetrain support	Mobility assemblies	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$3.00	\$3.00	Current	Allocated share of the same \$24.90 Technic assortment.
Structure	1	LEGO Technic Bushings / Spacers	Various	Axle positioning and alignment	Mobility assemblies	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$2.00	\$2.00	Current	Allocated share of the same \$24.90 Technic assortment.
Structure	1	LEGO Technic Connectors / Bracing	Various	Chassis, sensor and motor integration	Throughout chassis	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$4.00	\$4.00	Current	Allocated share of the same \$24.90 Technic assortment.
Mounting	1	Color Sensor Light-Isolation Casing	3D-printed/custom	Reduces surrounding-light influence	Around S4	Amazon (tracked material price)	PLA 1.75 mm, 1 kg spool	\$15.99	\$15.99	Current	Priced as one 1 kg PLA spool (\$15.99 Amazon tracked). Actual filament consumed by one casing is much lower.
Mounting	1	Color Sensor Mount	LEGO Technic/custom assembly	Holds S4 downward	Lower chassis	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$1.50	\$1.50	Current	Allocated share of the \$24.90 structural Technic assortment.
Mounting	1	HuskyLens Mount	LEGO Technic/custom assembly	Maintains forward camera orientation	Front/upper chassis	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$2.00	\$2.00	Current	Allocated share of the \$24.90 structural Technic assortment.
Mounting	1	Arduino Nano Support	Custom/LEGO mounting	Secures supporting controller	Internal/upper chassis	Amazon kit allocation	dsptwod 420pc kit / B0B4NDJRKZ	\$1.40	\$1.40	Current	Allocated share of the \$24.90 structural Technic assortment.
Wiring	2	EV3 Motor Cables	LEGO EV3 cables	Motor A and B connections	EV3 → Motors	Amazon/market cable-pack proxy	LEGO 45514 allocation	\$2.69	\$5.38	Current	Allocated from one EV3 7-cable pack reference value of \$16.12.
Wiring	4	EV3 Sensor Cables	LEGO EV3 cables	S1–S4 connections	EV3 → Sensors	Amazon/market cable-pack proxy	LEGO 45514 allocation	\$2.69	\$10.74	Current	Allocated from one EV3 7-cable pack reference value of \$16.12.
Wiring	1	Jumper / Connection Wires	ELEGOO 120pcs Dupont jumper wire kit, 20 cm	Vision subsystem integration	HuskyLens / Nano	Amazon (tracked)	ELEGOO EL-CP-004 / B01EV70C78	\$6.98	\$6.98	Current	120-piece Dupont jumper-wire kit; M-F, M-M and F-F.
Wiring	1	Cable Retention Elements	Various	Keep cables clear of moving mechanisms	Throughout chassis	Amazon (tracked)	100-pack 4-inch cable ties	\$0.99	\$0.99	Current	Current Amazon deal/reference for a 100-pack of 4-inch cable ties.

Complete BOM Pricing Summary

Priced line items	32
Unpriced line items	0
Complete replacement-cost estimate	\$940.18
Pricing status	ALL BOM ROWS PRICED